Aluminium-Water Combustion for Zero Carbon Marine Shipping: Masters Proposal

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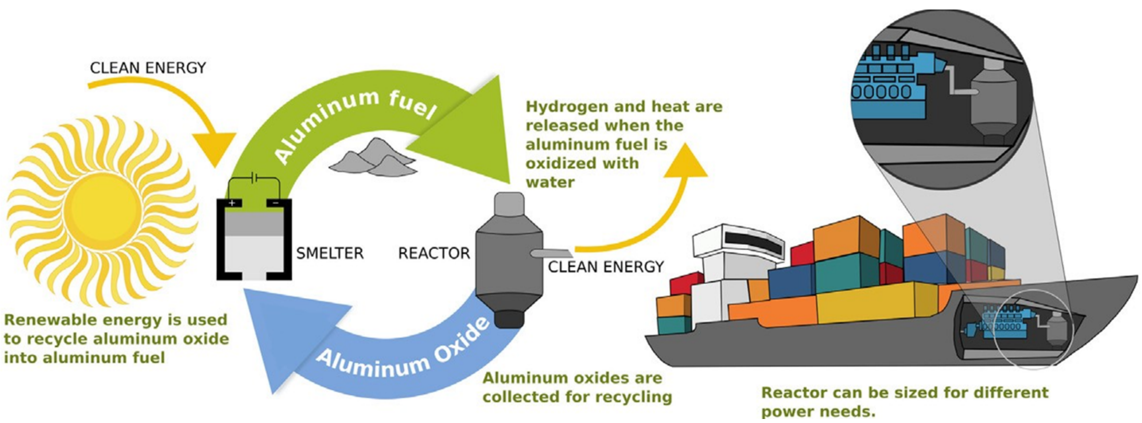

Figure 1: Aluminium powder as maritime shipping fuel (Energy Life-cycle). Adapted from [1]

Abstract

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1 Research Background

Overview

Maritime shipping plays a crucial role in global trade, facilitating around 80% of all commerce [7]. This makes it essential for globalization, economic development, and maintaining living standards. However, the sector also contributes significantly to global greenhouse gas emissions, accounting for 2-3% of global emissions and approximately 10% of all transport-related emissions [8, 7].

In response, the International Maritime Organization (IMO) has set ambitious targets to curb emissions from international shipping. By 2030, the IMO aims to reduce annual greenhouse gas emissions by 20-30% compared to 2008 levels, with a more aggressive target of 70-80% reduction by 2040 [9].

1.1 Preliminary Analysis of Electro-Fuels Options

Overview

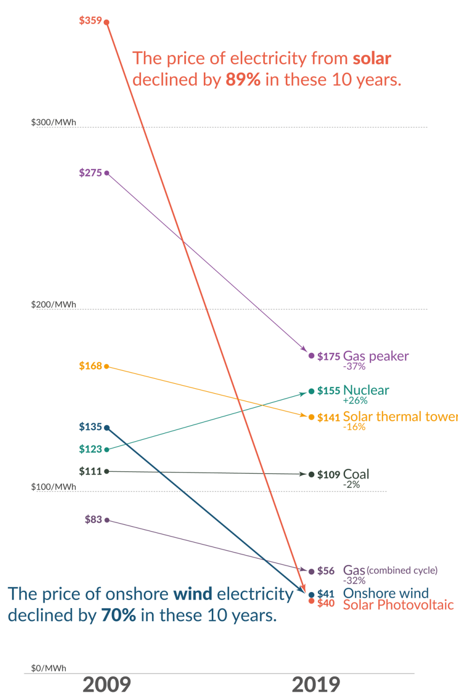

Figure 2: Falling cost of solar electricity vs Fossil Fuels [2]

Recent years have witnessed a dramatic decline in the cost of low-carbon electricity sources, particularly solar and wind power, as illustrated in Figure 2. This cost reduction has significantly enhanced the viability of transitioning to a low-carbon economy, yet substantial challenges persist in the so-called hard-to-abate sectors, among which marine shipping features prominently.

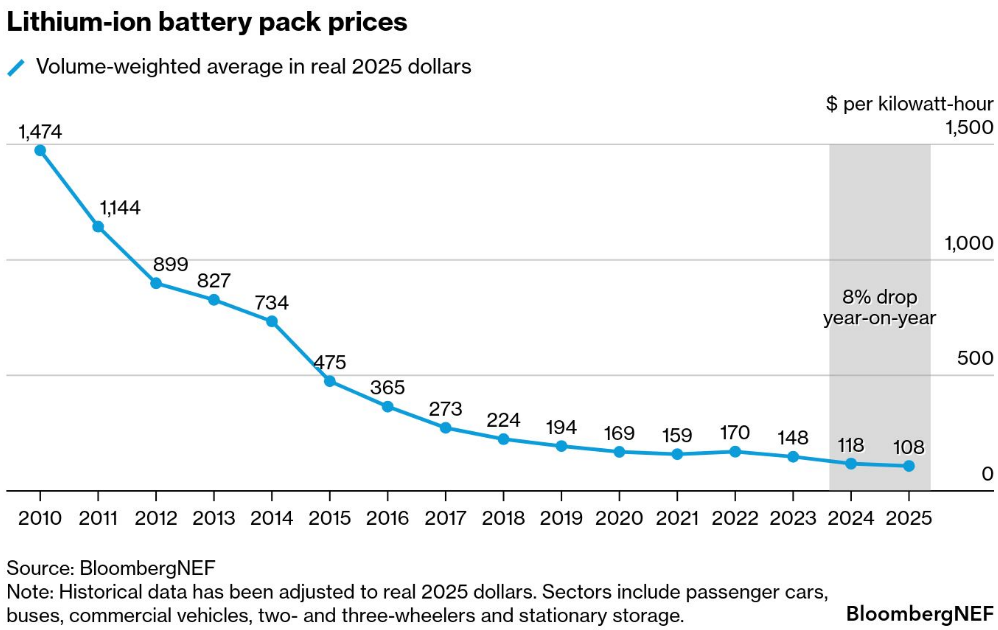

Figure 3: Falling cost of lithium ion battery packs [3]

The predominant method for storing electrical energy remains electrochemical batteries, with lithium-ion technology leading the charge. These batteries have increasingly displaced fossil fuels in passenger vehicles and, more recently, have begun penetrating short and medium-distance trucking applications [10]. This transformation has been driven largely by the precipitous decline in lithium battery prices, as demonstrated in Figure 3. The technology's reach has even extended to inland shipping routes, where standardized and swappable battery packs are being deployed in China [11]. Major battery manufacturers such as CATL are making substantial investments in these markets, signaling confidence in the technology's continued expansion [12].

However, despite remarkable advances in both price and energy density, lithium-ion batteries remain largely impractical for deep-sea shipping, which accounts for approximately 70% of marine shipping emissions [13]. The fundamental constraint lies in the current energy density limitations of lithium-ion technology, which ranges from 0.1 to 0.5 kWh/kg [14, 15]—substantially lower than gasoline, which can exceed 10 kWh/kg [16]. This considerable disparity in energy density presents a formidable barrier to the electrification of long-distance maritime transport.

In response to the energy density limitations of lithium-ion batteries, researchers and policymakers have increasingly explored the use of abundant and inexpensive solar electricity to produce alternative fuels, primarily through electrolysis. This approach has given rise to a diverse landscape of electro-fuel options, which can be systematically organized into four broad classifications:

- **Simple electro-fuels** - hydrogen and ammonia, which represent the most straightforward products of electrolytic processes

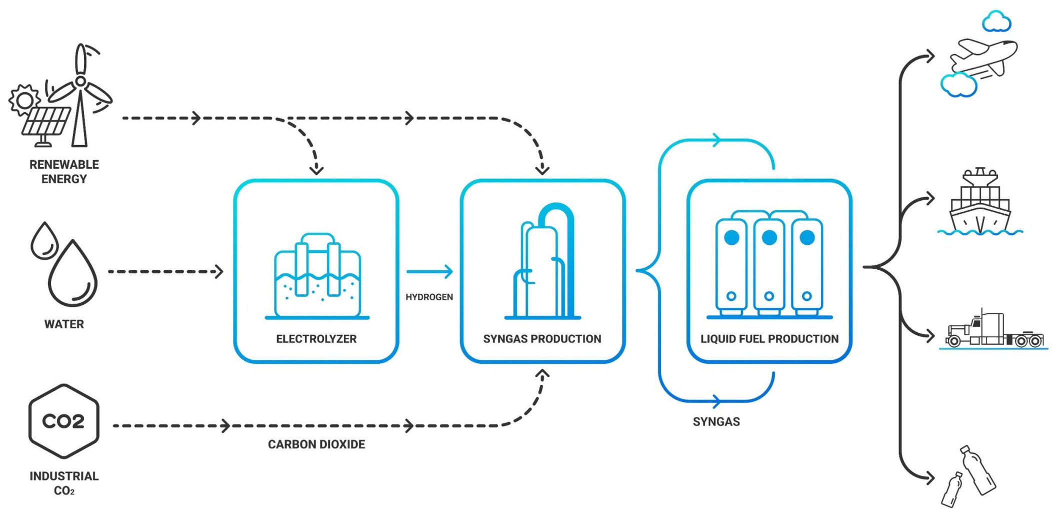

Figure 4: High Level Process Diagram for synthetic hydrocarbon production from renewable electricity [4]

- **Synthetic hydrocarbons** - primarily methane and methanol, whose production pathways are illustrated in Figure 4, which depicts the high-level process for synthesizing these fuels from low-carbon electricity

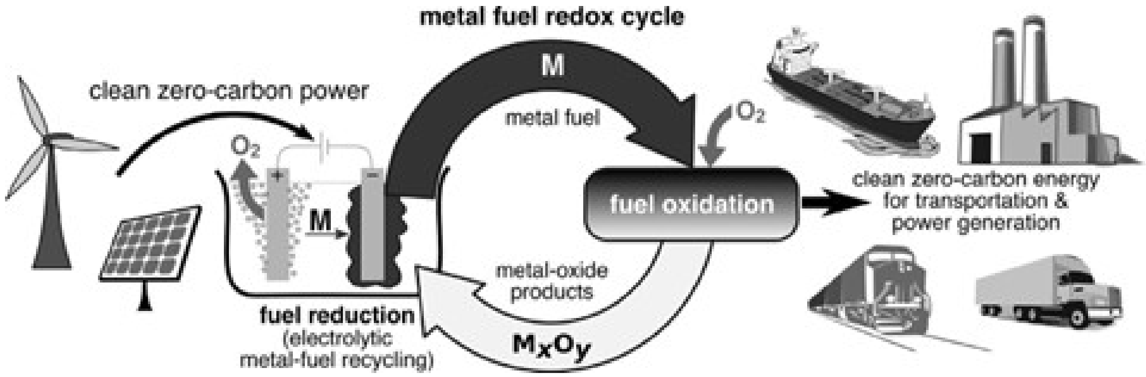

Figure 5: Metal based electro-fuels as secondary energy carriers [5]

- **Metal-based electro-fuels** - including iron, aluminium, magnesium, silicon, and boron. Figure 5 demonstrates the high-level process by which these metal-based electro-fuels can function as recyclable secondary energy carriers, offering a distinctive closed-loop energy storage paradigm

- **Metal hydrides (hydrogen carriers)** - such as MgH_2, AlH_3, $LiBH_4$, and $NaAlH_4$, providing an alternative means of storing and transporting hydrogen in a more dense and stable form than its gaseous state

In the following subsections, a preliminary analysis is conducted to compare the mass energy and volumetric energy densities of these various electro-fuel options in order to understand their viability for marine shipping applications.

1.1.1 High Level Energy Density Analysis

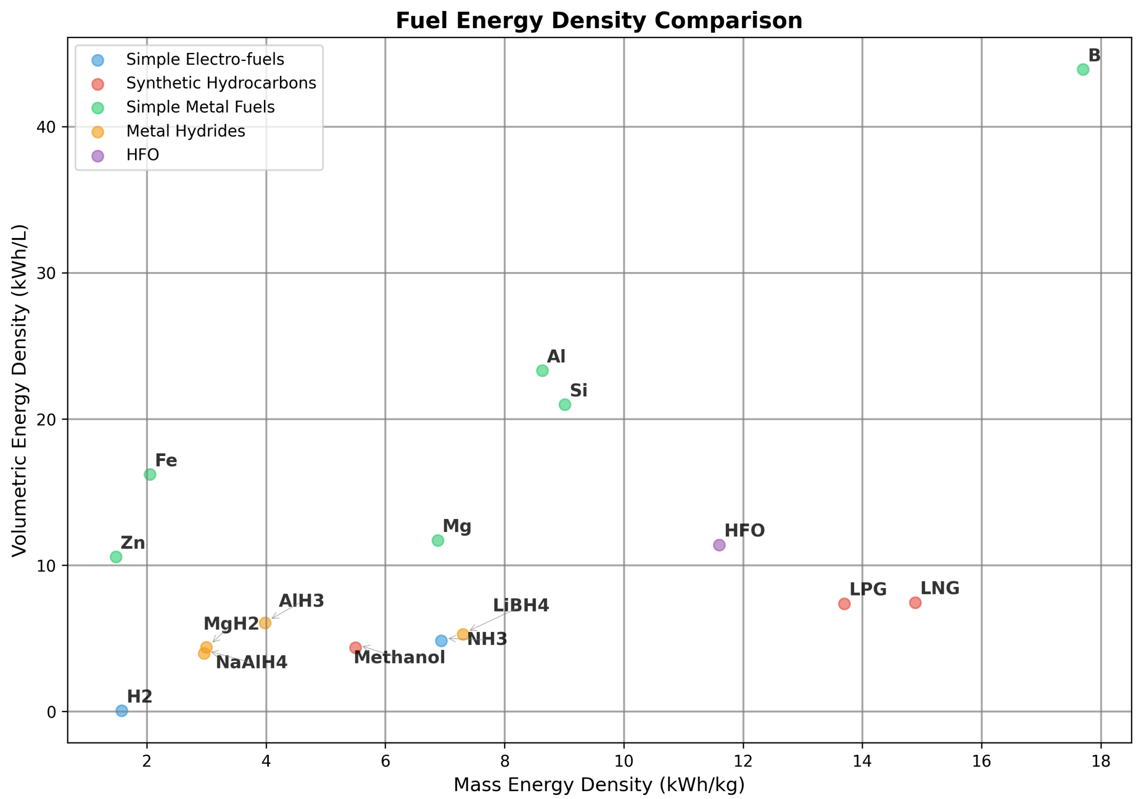

Figure 6: Simplified Energy Density Analysis. Data Analysis can be found in the Appendix. Adapted from [6]

2 Research Questions

3 Methodology

4 Proposed Project Timeline

5 References

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